

Dexterous Control: A Review

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Abstract

This note is intended to be general review of the current progress in building learning-based dexterous manipulation systems. In particular, we are primarily concerned with using neural networks to control “hand-like” manipulators.

1 Work Notes

Here is a non-exhaustive list of specific questions that I think are going to be important to study

- What are elementary and general purpose semantic action spaces that we should expect as *input* into a dexterous manipulator? “Open/close”?, Language?, Video?
- What should the low-level control output of a manipulator to physical hardware be?
- What is a list of all the sub-problems required to build a general purpose dex. manipulator? What can be built / solved entirely in simulation?

In order to understand what the state of the art is and where the current challenges are, we can compile a list of recent / seminal projects as representations of current methods.

From earlier reading, here is a list of relevant work.

- Eureka + DrEureka (Ma et al.)
- Learning Dexterous In-Hand Manipulation (OpenAI)